Reading the Drones: Working Towards a Critical Tradition of Interactive Poetry Generation

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Computer-generated poetry is now almost sixty years old, stretching from the work of Christopher Strachey, Jackson Mac Low and Theo Lutz in the 1950s to the wealth of programs freely available online today. As Antonio Roque notes in his essay 'Language Technology Enables a Poetics of Interactive Generation', it is a history without an 'overarching community', instead comprising four distinct but overlapping traditions that he enumerates thus:

- The Poetic tradition includes those who are interested primarily in the resulting poems: for [Jackson] Mac Low and the Gnoetry poets, the automated techniques being used are generally secondary to the poems being produced, and the program's output may be modified to produce an interesting poem. Most of the Digital Poets of the turn of the century may be classified here.
- The Oulipo tradition includes those who are interested in developing novel poetic forms, and in studying the types of poems that are generated from a given constraint or combinatorial method. To these poets, the authoring method being used is usually more interesting than the poem that it produced.
- The Programming tradition includes those aligned with the hacker sensibility, referring to those who enjoy understanding the workings of complex systems and finding creative and effective solutions to technical problems (Raymond 2004). Such programmers traditionally value the free distribution of resources such as programs, information, and infrastructure (Raymond 2011). JanusNode, ETC, and West's POETRY GENERATOR were distributed for the general good; Strachey's love letters, Dissociated Press, Travesty, and Mark V. Shaney were developed in computer labs principally for enjoyment.
- The Research tradition covers those interested in understanding language use and cognition (including critical work in the humanities) as well as those developing innovative techniques in language engineering. Examples are Lutz, Manurung, Gervás, and the NLP/CL/AI investigators of the past decade.[1]

[1] Antonio Roque, 'Language Technology Enables a Poetics of Interactive Generation' in The Journal of Electronic Publishing, 14:2, Fall 2011. Retrieved from <http://quod.lib.umich.edu/cgi/t/text/text-idx?c=jep;view=text;rgn=main;idno=3336451.0014.209> on 13th June 2013

This paper begins with the premise that despite the inherent literariness of computer-generated poetry, one 'tradition' is conspicuous by its absence: the Lit-Critical. While Roque pays lip-service to 'critical work in the humanities' in the Research tradition, the examples cited invariably come from the fields of computational linguistics and natural language processing. This is no deficiency of his typology, but rather points to the historical failure of literary criticism to engage seriously with computer-generated poetry. While occasional figures such as Charles O. Hartmann (whom Roque cites as a practitioner working across traditions) have brought their critical background to bear on their creative explorations of the mode[2], and the more recent work of Christopher Funkhouser has proved invaluable in sketching out a history of computer-generated poetry attentive to its literary context[3], the existence of a Lit-Critical tradition as such is doubtful, and its parameters remain ill-defined. It is the object of this paper to rectify this solecism, proposing a mode of critical engagement that might allow computer-generated poetry or, more specifically, the programs that enable it, to be naturalised as objects of textual study according to the protocols of literary criticism. Christian Bök has stated that 'the poets of tomorrow are likely to resemble programmers, exalted, not because they can write great poems, but because they can build a small drone out of words to write great poems for us.'[4] Bök – who has reportedly taught himself programming – sits squarely within the poetic tradition. This paper proposes a Lit-Critical analogue: we *read the drones*.

In order to achieve this, it will first give a working definition of the object of study, defined as an *interactive poetry generator*, suggesting some reasons why literary critics have failed to engage with such objects and why this failure is no longer tenable. Along the way, it will offer a selective history of the form that will inform the discussion that follows. It will then proceed from the *what* to the *how*, performing a comparative analysis between what might be construed as the first interactive poetry generator – Tristan Tzara's 'To make a dadaist poem' – and one of the most recent (and most powerful) – Chris Westbury's JanusNode. It will argue that a full critical understanding of Tzara's text can only proceed from a phenomenological engagement, attentive to what is termed the reader-plays-poet dynamic

[2] See Charles O. Hartmann, *The Virtual Muse: Experiments in Computer Poetry*, (New England: Wesleyan University Press, 1996)
[3] See Christopher Funkhouser, *Prehistoric Digital Poetry: An Archaeology of Forms 1959-1995*, (Tuscaloosa: University of Alabama Press, 2007) pp31-84
[4] Christian Bök , 'The Piecemeal Bard is Deconstructed: Notes Towards a Potential Robopoetics', Object 10: Cyberpoetics (2002) p17 Retrieved from Ubuweb <http://www.ubu.com/papers/object/03_bok.pdf> on 6th September 2012

that is a feature of any Dadaist poem. This approach will then be applied to present-day interactive poetry generators via an interface-centred close reading of JanusNode that draws on the phenomenology of Gaston Bachelard and the work of concrete poet Eugen Gomringer. This analysis will serve to assert the literary pedigree of interactive poetry generation as well as establish some ways to critically fix a textual object for which flux is a primary characteristic. Finally, the paper will briefly address *why* literary critics should study interactive poetry generators, referencing the work of Kenneth Goldsmith to show how these programs offer a means to develop and augment discussions of contemporary poetics. Throughout, this paper strives to encourage deeper critical engagement with computer-generated poetry and the recognition that interactive poetry generators constitute virtual aesthetic objects in their own right worthy of literary study. Furthermore, it aims to engage Roque's other 'traditions' in dialogue, in the hope of further developing and extending the myriad possibilities of the form.

1. WHAT?

Among the most powerful poetry generators available online today is JanusNode, a freeware program by Chris Westbury. The manual describes it as 'a very simple virtual robot built to fulfil a single function: to manufacture textual possibility (known to some as: nonsense) that material entities such as yourself endow with value.'[5] More conventionally, and following Roque's terminology, it might be termed an *interactive poetry generator* (or IPG). While the term is used advisedly (see below), it nevertheless succinctly addresses the processes involved in using the program. The *generator* component is that 'manufacture [of] textual possibility' – the raw text that constitutes the program's output. To become *poetry*, however, it requires a 'material entit[y to] endow [it] with value' – a reader who discovers poetic images and turns of phrase in that output. But so far this is only a one way process, hence the *interactive* component. Using the interface, the reader is able to modify the output in various ways both before and after generation. Thus do we arrive at the model of dynamic engagement shown in Figure 1: the program responds mechanically to our input as we respond imaginatively to its output. According to the manual, JanusNode operates on a principle of 'Divine Indifference'[6], the opposite of divine intervention. That is, whenever we discover 'meaning' in JanusNode's output, whether banal, hilarious, profound or otherwise, it

[5] Chris Westbury, *JanusNode 3.12 Documentation* (hereafter referred to as *JND*), 2012 p2
[6] *JND* p4

is always us who put it there. This is a creative act; it is not a critical one. But as this paper argues, the possibility of *critical* engagement with IPGs becomes real if we look beyond the *poetry* component – those results that we subjectively imbue with value – to the *interactive* component which, as can be seen, encompasses the others in its loop. In the terms of literary analysis, this requires a phenomenological approach. We have to read ourselves reading or, more accurately, *using* or *playing*.

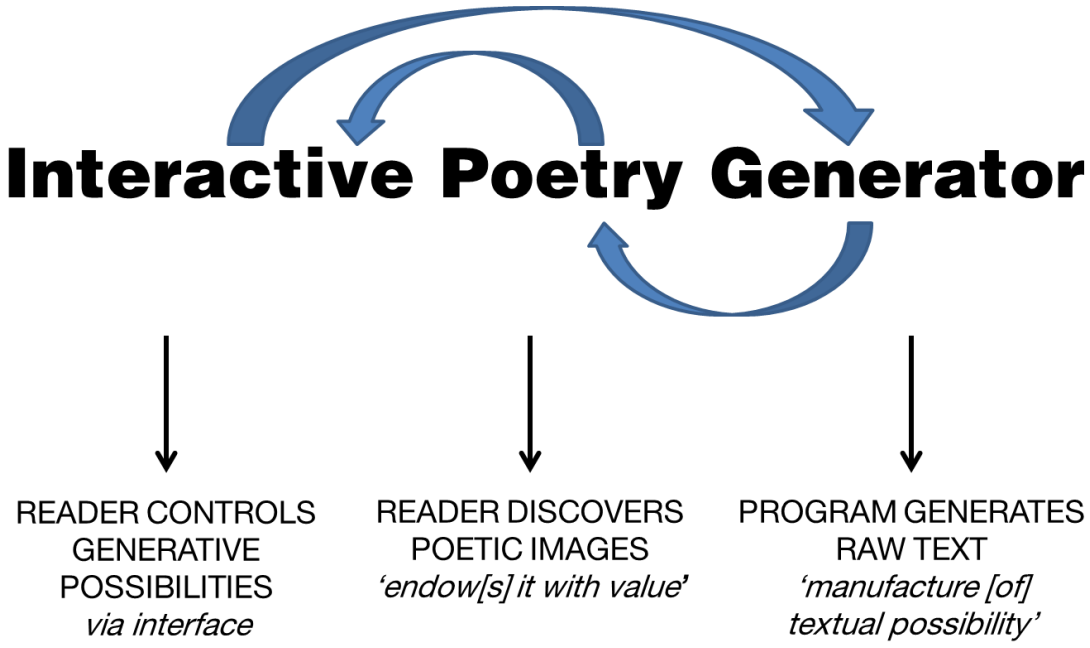


Figure 1.

As Funkhouser and Roque note, the first computerised poetry generator was written by Informatics student Theo Lutz in 1959, generating lyric poetry using words and structures from Kafka's 'The Castle'[7]. Its title, 'Stochastic Text', is telling. Until computers are developed enough to engage with language on a deep semantic level, all computer generated text will have a random element. Certainly, JanusNode works on a similar, albeit more advanced, principle of aleatoric selection and combination. But Lutz's program cannot be considered an interactive poetry generator, as it lacks one crucial prerequisite: interactivity.

In a discussion of interactive fiction in *Cybertext: Perspectives on Ergodic Literature*, Espen J. Aarseth criticises the term 'interactive' as either meaningless or trivial. Since a non-trivial

[7] Theo Lutz, 'Stochastic Text' (1959), quoted in C.T. Funkhouser, *Prehistoric Digital Poetry: An Archaeology of Forms, 1959-1995*, University of Alabama Press: 2007, Kindle ed.

definition of the word implies 'a functional equality between the interacting agents' – that is, human and machine – and that a functional equality can only exist 'if the machine is somehow aware of the situation', which remains impossible, it follows that interactive fiction is no more interactive than conventional fiction, insofar as a 'successful fiction must […] in one sense be interactive, just as a lie needs a believer in order to work.'[8] One might add that a poem needs a reader in order to be understood and enjoyed, and such is the case too with the interactive poetry generator. The crucial difference is that if we assume that poets can understand and enjoy their own work, then the output of an IPG needs a reader to be understood and enjoyed *at all*. So while the 'trivial' interactivity of IPGs is, as Aarseth suggests, 'perhaps better described as participation, play, or even use,'[9] it is a play that requires not only a cognitive engagement but also a physical one – not just reading, but *doing*. Hence Lutz's text cannot be described as interactive insofar as the program itself remains inaccessible to the reader/doer, whether through lack of suitable channels of distribution or of a user interface that requires no specialised technological knowledge.

As such, the first interactive poetry generator proper precedes Lutz's program by forty years and falls into none of Roque's traditions (except, necessarily, the Poetic). It is rather a key text of experimental modernism, Tristan Tzara's 'To make a dadaist poem':

> Take a newspaper.
> Take a pair of scissors.
> Choose an article as long as you are planning to make your poem.
> Cut out the article.
> Then cut out each of the words that make up this article and put them in a bag.
> Shake it gently.
> Then take out the scraps one after the other in the order in which they left the bag.
> Copy conscientiously.
> The poem will be like you.
> And here you are a writer, infinitely original and endowed with a sensibility that is charming though beyond the understanding of the vulgar. [10]

[8] Espen J. Aarseth, *Cybertext: Perspectives on Ergodic Literature*, (Baltimore: John Hopkins Press, 1997) p50
[9] Aarseth p49
[10] Tristan Tzara, 'To make a dadaist poem'(1920, pub. 1924), quoted in Robert Motherwell (ed.), *The Dada Painters and Poets: An Anthology (2nd ed.)*, Belknap/Harvard University Press: 1951; 1979 (repr. 1981) p.92

This text will be examined in more detail below, but in the meantime it is enough to state that the text invites the construction of an interface-like space in which the reader must engage physically with the 'program' in order to produce the desired result; namely, an entirely new poem. It is therefore arguably closer in kind to JanusNode than 'Stochastic Text' and remains the best literary model against which to read contemporary IPGs.

According to this logic, the next major development in interactive poetry generation is Brion Gysin's cut-up method, developed in the 1950s and popularised by William Burroughs. If the Dadaist poem is construed as an agrarian method of poetry generation, then the cut-up constitutes an industrial revolution, allowing for the production of chance juxtapositions with brutal efficiency (in contrast to the laborious operations of Tzara's text). Like the Dadaist poem, the materials for its creation are widely and immediately available, as is the 'program' itself, described several times throughout its most famous output, Burroughs's *Nova Trilogy*. The ease with which the reader/doer can access and produce these texts has more in common with contemporary IPGs than Lutz's program, for which engagement is only possible through a sample output.

The first flickerings of a digital revolution occur in 1985 with the publication of 'the first book ever written by computer'[11]: *The Policeman's Beard is Half-Constructed* by the text generation system RACTER. Notably, this book was published not by a poetry press but by one specialising in technical computer titles; more notably still, a scaled-down version of the program was also briefly commercially available. Here, the interface is no longer one of scissors and Stanley knives but pixels and a keyboard. In principle, this program brought the *digital* interactive poetry generator outside the exclusive province of literarily-minded programmers and technophile poets. Like avant-garde poetry books, however, such programs are rarely profitable, so it is only with the advent of the web that the digital revolution of IPGs truly arrives. Today, there are numerous programs available online for free download (one directory – by no means comprehensive – lists over two dozen[12]). Advanced text generation is now accessible to all. In such an environment, the failure of the literary critic to engage with IPGs is no longer tenable. But how can this failure be overcome?

[11] RACTER, *The Policeman's Beard is Half-Constructed: computer prose and poetry by Racter. (The First Book Ever Written by a Computer).* Warner Books: 1985

[12] edde addad, 'Interactive Poetry Generation Systems: An Illustrated Overview'(2010). Retrieved from netpoetic.com <http://www.netpoetic.com/2010/10/interactive-poetry-generation-systems-an-illustrated-overview/> on 11th September 2012

2. How?

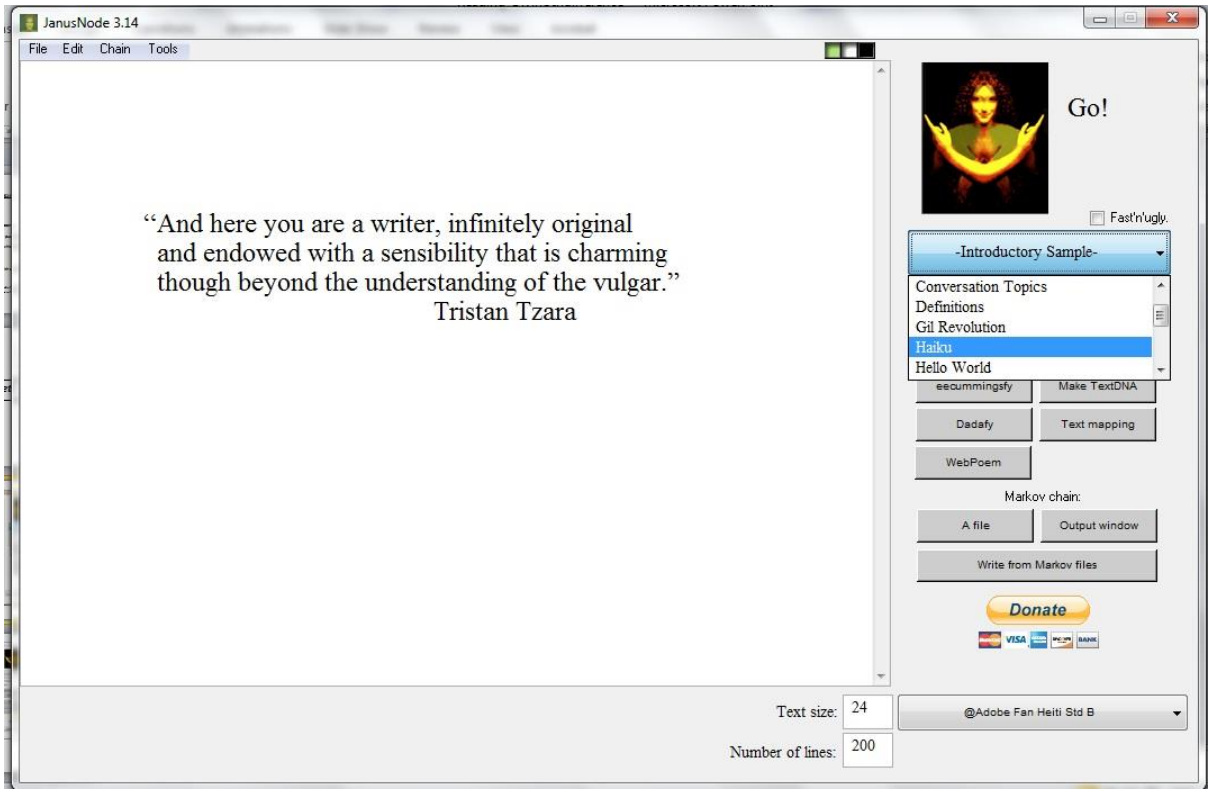


Figure 2.

When we first open JanusNode, we are presented with an epigraph, the final line of 'To make a dadaist poem' (Figure 2). This serves as our point of departure: as noted above, although it requires a human to perform its mechanical operations, it is in effect the first IPG. By reading Tzara's text, then, we might learn how to read JanusNode. 'To make a dadaist poem' presents three possible approaches. We could study a sample output, but as a product of 'Divine Indifference', this would constitute a creative act, not a critical one. We could study the instructional text itself, treating it as a kind of meta-poem. This certainly has its uses, but is akin to studying respiration in a cadaver. The Dadaist poem invites the reader not just to read, but to *do*. The comprehensibility and value of both output and instructions are contingent upon the activity itself. As such, it is necessary to take a phenomenological approach: we have to play Tzara's game.

When we do so, the first thing we realise is how time-consuming it all is: cutting up even a short article is a precise and boring task. This combination of mechanical attentiveness and

intellectual disengagement gives the task a meditative quality. Like any repetitive physical labour, it encourages daydreaming, which may or may not be triggered by the suddenly rootless signifiers in hand. But the Dadaist poem is a game of two halves. It pivots on the volta-like 'Shake it gently'. As we take the words out one by one our daydreaming focuses on the emergent text. Long before the poem is finished we are reading 'meaning' in its output, fulfilled and denied by turns with each successive element we reveal. Crucially, this mode of engagement is predicated on the instruction 'copy conscientiously' – the only instruction that cannot be performed by a robot (a robot, of course, is without conscience). Hence the gentle shake engineers an ontological shift: from reader to writer; consumer to producer; robot to poet. Tzara's text is often considered a rhetorical gesture, a negation of culture concerned only with short circuiting modes of everyday discourse. But this reading fails to acknowledge the dynamic at work whenever a reader is prompted to reach for the scissors. 'To make a dadaist poem' is an assertion of activity over passivity, creative play over consumption, situation over spectacle. This assertion is only made clear, however, if we consent to play 'poet' and follow Tzara's instructions.

The process involved in Tzara's text follows this formula: *reader-plays-poet*. We begin as readers and, if we choose to play the game, become poets, 'infinitely original' in Tzara's words. There is a parallel here with the dictum of Language poetry, that avant-garde movement in American poetry of the latter half of the 20th century, which may be expressed as reader-constructs-meaning. But here it is taken to its ludic extreme – a limit-case of Roland Barthes's writerly text. It is important to note that although 'reader' precedes 'poet', neither term is dominant, as both are consolidated in the idea of 'player'. Much as a player of a musical instrument is both reader (of the music) and performer (of the song), the maker of a Dadaist poem must both follow the signifier and perform the signified, otherwise the poem cannot exist.

Therefore, in order to read JanusNode and IPGs like it, it is necessary to take a phenomenological approach that focuses on the ways in which reader-plays-poet. Hence, we should focus not on specific outputs, but general structures, trends and possibilities. We should examine the ways in which this output is presented and – crucially – the ways in which it can be controlled and edited by the user. Though we should attend to the rhetoric of the accompanying documentation as we would the critical writings of a poet, the locus of our

investigation should be the interface itself and the mode of engagement it encourages. It is now possible to turn to JanusNode to demonstrate how this might be done.

Where Tzara would have us choose a newspaper article, JanusNode offers a whole array of possibilities, such as 'Definitions', 'Conversation Topics' and 'Haiku'. The user clicks 'Go!' and the screen fills with endless poetry (or rather, as many lines as the user dictates in the box on the bottom right). JanusNode produces the text quicker than we can read, so we see the text being generated before we are able to read it. Once generation has finished, however, poetry begins.

In order to read ourselves reading 'Divine Indifference' this paper draws on the work of Gaston Bachelard. Best known as a philosopher of science, Bachelard's most obvious contribution to literary theory has been his notion of the epistemological rupture which, refracted through the work of Louis Althusser and Michel Foucault, informs postmodern scepticism towards metanarratives. But in tandem with this work he also produced a phenomenological philosophy of poetry and the imagination, most fully realised in *The Poetics of Space.* There are three aspects of this work that, while perhaps making it resistant to conventional literary criticism, make it particularly amenable for reading JanusNode. The first is his emphasis on the isolated poetic image to the exclusion of pretty much everything else. In *The Poetics of Space*, he states:

> One must be receptive, receptive to the image at the moment it appears: if there be a philosophy of poetry, it must appear and re-appear through a significant verse, in total adherence to an isolated image; to be exact, in the very ecstasy of the newness of the image. The poetic image is a sudden salience on the surface of the psyche […][13]

It needn't be added that this presents a serious obstacle to reading anything longer than a short lyric or haiku. But it is on the level of the isolated image that JanusNode operates best. So as the maker of a Dadaist poem must be 'conscientious', the reader of JanusNode must be 'receptive to the image at the moment it appears'. How? As if we are panning for gold in a virtual Heraclitean river, we sift through the results, our cursor hovering over the scroll down

[13] Gaston Bachelard, *The Poetics of Space*, (Boston: Beacon Press, 1964, repr. 1994) p.xv

button, until something glitters – a result gives us pause. But whereas the glitter of gold is an objective phenomenon, the glitter of poetry is a subjective process of the active imagination. The randomly generated combination of signifiers resonates poetically; we experience that 'salience on the surface of the psyche'; we go 'wow' or 'oooh' or 'cool', and probe deeper into its 'meaning', supplying the text with imaginative content. Bachelard calls it *valorisation*, defined by his translator Colette Gaudin as 'the spontaneous activity of imagination attributing subjective values to its objects'[14]. As we respond imaginatively to items of JanusNode's output, we imbue them with the value of poetry. Figure 3 shows one such object that caught the imagination of the present author.

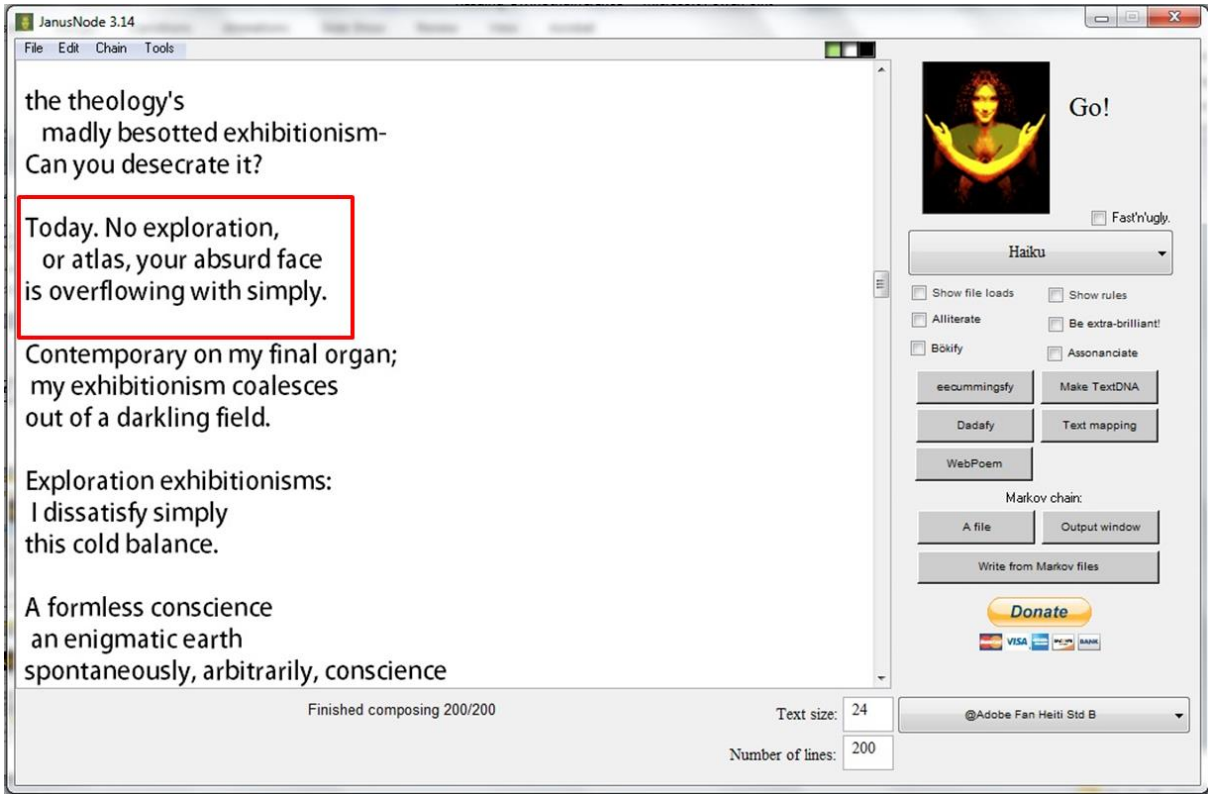


Figure 3.

Hence the second reason: for Bachelard, being 'receptive to the image' entails a strategy of reading antithetical to that of the literary critic. While the critic's reading is 'necessarily severe', the phenomenologist must begin in admiration. 'We can admire more or less,' says Bachelard, 'but a sincere impulse, a little impulse towards admiration, is always necessary if we are to receive the phenomenological benefit of a poetic image.'[15] To an extent, computer-generated literature already encourages this strategy of reading. A reader who is well

[14] Gaston Bachelard, *On Poetic Imagination and Reverie (Revised Edition)*, trans. and ed. by Colette Gaudin, (Dallas: Spring Publications, 1987, repr. 1998) p5
[15] *The Poetics of Space* p.xxxvi

disposed towards technology is likely to be more forgiving of a computer-generated text than one composed by a human author, inasmuch as stylistic criticisms are muted by a certain 'awesome factor' – the amazing fact that a computer can be programmed to write something imaginatively resonant at all. Ultimately, what we're marvelling at here is not the technology as such, which at any rate isn't particularly advanced, but the power of language itself to *mean*. 'Language dreams'[16], says Bachelard; it is remarkable that it does so even with no human point of origin. The aleatory language of Dada, Burroughs, RACTER and JanusNode exhibits a surprising degree of imaginative inhabitability – but only for those with that 'sincere impulse […] towards admiration.'

For Bachelard, this impulse is the consequence of another; namely, that readers of poetry dream of being poets. He states: 'in reading we are reliving our temptations to become a poet. All readers who have a certain passion for reading, nurture and repress, through reading, the desire to become a writer.'[17] So our admiration for and imaginative responses to poetic language are themselves the result of what is at root a deep *creative* drive. Hence the third reason why Bachelard is useful: if his statement can be expressed in the formula reader-dreams-of-being-poet, then the reader-plays-poet dynamic of JanusNode presents a means of erasing the distance between these two positions – from dream to (virtual) reality. It achieves this via the interface.

JanusNode features a variety of scripts, but its controls are not limited to this jukebox selection. Prior to generation checkboxes allow the user to employ a number of poetic devices, including assonanciate, alliterate, and Bökify (named after the poet, this introduces a lipogrammatic constraint). After generation, further manipulation is possible, either manually or using various buttons that include Dadafy (an emulation of Tzara's game), eecummingsfy (which breaks the words down into parts a la e.e. cummings) and Text mapping (which 'translates' the text into various idioms including postmodern, leetspeak and dubyabushism). There are more advanced functions too, such as the option to randomly generate a poem using text harvested from specified (or random) URLs and Wikipedia articles. The interface, then, presents a sandbox laboratory wherein reader-plays-poet, testing out various possibilities and measuring the results. According to the extended definition of interactivity provided above, we work on the text imaginatively *and* actively. While bound by the

[16] *The Poetics of Space* p146
[17] *The Poetics of Space* p.xxxvi

functions and limits of the interface (we are not *completely* free), this degree of active control heightens imaginative engagement, as we literally 'play' the texts that are generated before us.

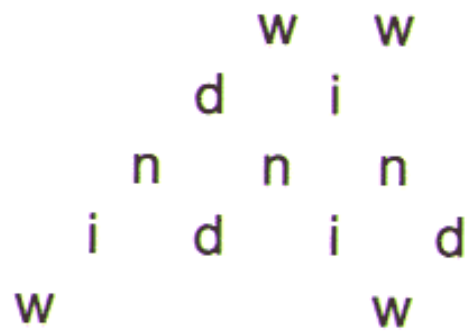


Figure 4.

Before concluding this analysis it is useful to turn briefly to the concrete poetry of Eugen Gomringer. Figure 4 shows 'wind'[18], an example of what Gomringer called the 'constellation'. A constellation, he says, is 'a play-area of fixed dimensions […] ordered by the poet. He determines the play-area, the field or force and suggests its possibilities. The reader, the new reader, grasps the idea of play and joins in.'[19] Reading 'wind', we follow the letters however we wish. Soon, the movement of our eyes becomes a kind of active metaphor for the movement of the wind – in this sense, the reader 'plays' the text, as we respond cognitively to our own physical engagement. If we continue reading we soon approach semantic saturation, whereby the poem becomes less a vehicle of meaning as a material object in its own right. JanusNode presents a virtual field of imaginative possibility that can be construed as a kind of advanced textual 'play-area'. Though ultimately 'ordered by the [programmer]', it depends to an unprecedented degree on a reader who 'grasps the idea of play and join[s] in'. And much as Gomringer's poem is designed to emphasise its *materiality*, the interface of JanusNode has a function designed solely to emphasise its *playability*: the 'Be extra-brilliant!' checkbox (Figure 5).



Figure 5.

[18] Eugen Gomring, 'wind' (c.1954). Retrieved from UbuWeb <http://www.ubu.com/historical/gomringer/gomringer03.html> on 11th September 2012
[19] Eugen Gomringer, 'From Line to Constellation' (1954). Retrieved from UbuWeb <http://www.ubu.com/papers/gomringer01.html> on 11th September 2012

As the manual tells us, this button is 'a placebo. It doesn't actually have any effect but you can probably fool yourself into believing that it does, and that will make you happy.'[20] Like the concrete poem, which Gomringer considered a 'functional object'[21], the 'be extra-brilliant' button is functional, but like the concrete poem, its function is entirely aesthetic and imaginative. It basically purports to be a turbo button for the imagination. And although the manual concedes that it is only a placebo, it serves to emphasise exactly the processes at work in JanusNode. It reminds us that we must 'be receptive to the image [or to brilliance] at the moment it appears' and to read with that 'sincere impulse towards admiration' (that is, if we believe, or affect to believe, that it works, and 'imagine harder' accordingly, we will be rewarded). What is more, by embedding this reminder in the interface itself, it affirms these imaginative processes as intrinsic to the whole enterprise of poetry generation, and establishes the interface as the locus of this activity. In short, it embodies the reader-plays-poet dynamic.

All IPGs participate in this dynamic, although the ways in which they do so varies. For example, another popular IPG, Gnoetry, limits its 'play-area' to a single poem, whose contents the user is able to regenerate in a process reminiscent of the dice game Yahtzee. For this reason other IPGs may well demand different frameworks through which to be read, but these readings should at least share the interface-centred, phenomenological approach that is proposed here, concerned with elucidating the mechanisms of the reader-plays-poet dynamic. While on the surface this may seem like a major departure from the study of static poems in a book, it is hoped that this analysis has shown how the theoretical toolkit of literary criticism is well equipped to make this shift and, moreover, that for those used to working with experimental and avant-garde texts, it is not such a departure after all.

One final question remains, however: how do we know the game is finished? Put differently, is there any ontological distinction between the reader-playing-poet and a real poet? Some might consider this a moot point, but it is at least practically useful to localise the point at which the game – as defined by the IPG in question – ends. With Tzara it is easy: when the bag is empty. Likewise is it simple enough to define the game limits of JanusNode. Its output

[20] *JND* pp37-8
[21] Eugen Gomringer, 'The Poem as Functional Object' (1960). Retrieved from UbuWeb <http://www.ubu.com/papers/gomringer04.html> on 11th September 2012

window, although it allows manual editing, is a volatile environment. It has no redo function, and one wrong click can all too easily wipe it clean. Hence any choice nuggets we particularly want to keep must be saved and transferred into a word processor for more elaborate editing and, ultimately, posterity. Once the text is shifted from the playful JanusNode interface to the sober environment of the word processor, it is game over. The good news, of course, is that the game is won: we are now poets.

3. WHY?

Of course, things are not as simple as the closing rhetoric of the previous section makes out. In terms of the close reading presented above, it makes sense to equate the limits of the 'text' under study with the limits of the interface, but the work of the literary critic does not end here. The ontological distinction between the reader-playing-poet and the poet, while localisable in terms of activity, still requires a great deal of theoretical interrogation. Furthermore, if it is the case that when the game ends, the poem begins, it is necessary to emphasise that the study of IPGs should not be at the expense of poets who use them. Some of the most interesting contemporary poetry is written in collaboration with IPGs and is yet to receive the critical attention it deserves – see, for example, the Gnoetry Daily group blog and Eric Elshtain's Beard of Bees press. Ultimately, however, once removed from the interface and the reader-plays-poet dynamic, such poetry must survive (or otherwise) on its own merits. But as this paper has argued, IPGs constitute virtual aesthetic objects in their own right which, though not without their precursors in the literary avant-garde, engender entirely new modes of poetic engagement. That these programs are usually hobbyist projects, bearing the idiosyncratic marks of their creators, only make them more conducive to close and comparative analyses. Moreover, they participate in a counter-tradition of extending the limits of poetic possibility that goes back at least as far as Dada. For these reasons alone they are worthy of literary study. Not only that, but as contemporary discourse around poetry centres on questions of appropriation and authorship in a virtual culture of information overload and archive fever, it can be argued that such work has never been more relevant. How, then, can IPGs and their study contribute to this discourse?

Among contemporary poets, few are as engaged with the discourses of technology and new media as conceptual poet and/or *un*creative writer Kenneth Goldsmith, as the title of his critical work, *Uncreative Writing: Managing Language in the Digital Age*, suggests. There is

a Futurist ecstasy running through Goldsmith's rhetoric, centred on the increased possibilities for the manipulation of text presented by new technology:

> The writer's solitary lair is transformed into a networked alchemical laboratory, dedicated to the brute physicality of textual transference. The sensuality of copying gigabytes from one drive to another: the whirr of the drive, the churn of intellectual matter manifested as sound. The carnal excitement from supercomputing heat generated in the service of literature. […] Language in play. Language out of play. Language frozen. Language melted.
>
> Sculpting with text.
>
> Data mining.
>
> Sucking on words.
>
> Our task is simply to mind the machines.[22]

Is Goldsmith describing something akin to the reader-plays-poet dynamic? Not quite. He exhibits the 'sincere impulse towards admiration' advocated by Bachelard, but it is less admiration for the poetic image – the 'literature' *per se* – as it is for the 'supercomputing heat', the technology itself – not the image, but the 'brute […] transference'. In other words, Goldsmith's rhetoric, much like his work as an uncreative writer, fetishizes the process at the expense of the product. This may seem to tally with the interface-centred phenomenological approach advocated in the previous section. But Goldsmith generally publishes his work in traditional book or pdf format, therefore inviting a traditional mode of critical engagement (such as Marjorie Perloff's close reading of Goldsmith's *Traffic* in *Unoriginal Genius: Poetry by Other Means in the New Century*[23]), so that any dynamic mode of engagement is closed to the reader. The result is a series of books that are unashamedly and almost unbearably dull, such as his infamous work *Day*, a retyping of one edition of the New York Times. Goldsmith wears this dullness as a badge of pride, boasting that he is 'the most boring writer that ever lived'[24] and that he prefers a 'thinkership' to a readership. In terms of interactivity his work travels in the opposite direction to IPGs, reducing participation to the *purely* cognitive as IPGs augment the cognitive with the

[22] Kenneth Goldsmith, *Uncreative Writing: Managing Language in the Digital Age*, (New York: Columbia University Press, 2011) p221

[23] Marjorie Perloff, *Unoriginal Genius: Poetry by Other Means in the New Century*, (Chicago: University of Chicago Press, 2010) pp146-66

[24] Kenneth Goldsmith, 'Being Boring' (2004), retrieved from Buffalo Electronic Poetry Center <http://epc.buffalo.edu/authors/goldsmith/goldsmith_boring.html> on 14th June 2013

physical. While both uncreative writing and IPGs emphasise process and procedure, the latter makes the reader an instrumental component of this process. As such, it is a more imaginatively proactive experience for the reader, serving as a useful critical corrective to the more problematic aspects of Goldsmith's work and the Conceptual Writing movement in general.

Moreover, if the implicit point of a work such as *Day* is that the glut of information in the digital age nullifies any scope for originality and creativity, then are we not better, polemically speaking, to renounce our ties to 'information' in the conventional sense and bask in the ludic space of the reader-plays-poet dynamic? A space where the poetic image reigns supreme, and where our 'temptations to become a poet' are not reduced to the mechanistic procedures of a dry Conceptualism such as Goldsmith's but are realised in an aleatoric and engaging virtual environment. Goldsmith is fond of quoting John Cage's quip that 'technology essentially is a way of getting more done with less effort'[25]. The IPG suggests a shift in emphasis that might rescue poetry from the void of the boring yet: technology is essentially a way of *having more fun* with less effort.

William Carlos Williams famously said that 'a poem is a small (or large) machine made of words'[26]. The conceptual writing of Goldsmith and others asserts the truth of this statement in its most literal sense, composing their works according to procedural constraints that might be construed as specially designed 'programs'. But in so doing such writers forget another truism of Williams's: 'if it ain't a pleasure, it ain't a poem'[27]. Williams's joyous celebration of the imagination in *Spring and All* has its contemporary echo in the reader-plays-poet dynamic of the interactive poetry generator, whose operations are wholly dependent on the active imagination of the user. *We* give it meaning, *we* find the poetry, and *we* bring the joy. Nor is this an entirely frivolous activity, so long as we have the capacity to experience that 'sudden salience on the surface of the psyche'. There is a passage in *Spring and All* that seems to speak of this condition which, with a small alteration, may well be spoken by JanusNode itself, describing the very essence of the reader-plays-poet dynamic:

[25] John Cage, quoted in *Uncreative Writing* p82
[26] William Carlos Williams, quoted in 'Introduction to *The Wedge*', Poetry Foundation, 2009. Retrieved from < http://www.poetryfoundation.org/learning/essay/237888> on 14th June 2013
[27] William Carlos Williams, Reading and Commentary at Harvard University, December 4, 1951. Retrieved from PennSound < http://media.sas.upenn.edu/pennsound/authors/Williams-WC/13_Harvard_12-04-51/Williams-WC_01_Commentary-1_Harvard_12-04-51.mp3> on 14th June 2013

> And if when I pompously announce that I am addressed – To the imagination – you believe that I thus divorce myself from life and so defeat my own end, I reply: To refine, to clarify, to intensify that eternal moment in which we alone live there is but a single force – the imagination. This is its book. I myself invite you to read and see.
>
> In the imagination, we are from henceforth (so long as you read) locked in a [virtual] embrace, the classic caress of [interface] and [reader-playing-poet]. We are one. Whenever I say "I" I mean also "you." And so, together, as one, we shall begin.[28]

It is thus that the role of IPGs in contemporary poetry should be considered: as a means of retaining the imagination as fundamental to poetry even as discussions tend towards ideological, technological and paratextual concerns, whilst still participating in those discussions. In other words, IPGs serve as a hedonistic antidote to the driest excesses of Conceptual Writing and its ilk without losing sight of the pertinent critical issues raised by that work. As literary critics, it is our duty to accommodate 'texts' like JanusNode in such discussions, and to help foster appreciation of them and bring them to a wider audience. It is hoped that this paper has offered some tentative early steps as to how this might be achieved.

[28] William Carlos Williams, *Spring and All* (1923), reprinted in *The William Carlos Williams Reader*, ed. by Macha Louis Rosenthal (New York: James Laughlin, 1966) p178

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